

Unit 12, Lesson 1: Solving Problems with Lengths of Intersecting Chords



Benchmark

MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.



Learning Target

- ☐ I can solve mathematical and real-world problems involving lengths formed by the intersection of two chords.



12.1.1 Warm-Up: Notice and Wonder

1. A picture of a bowling-ball pyramid is shown.

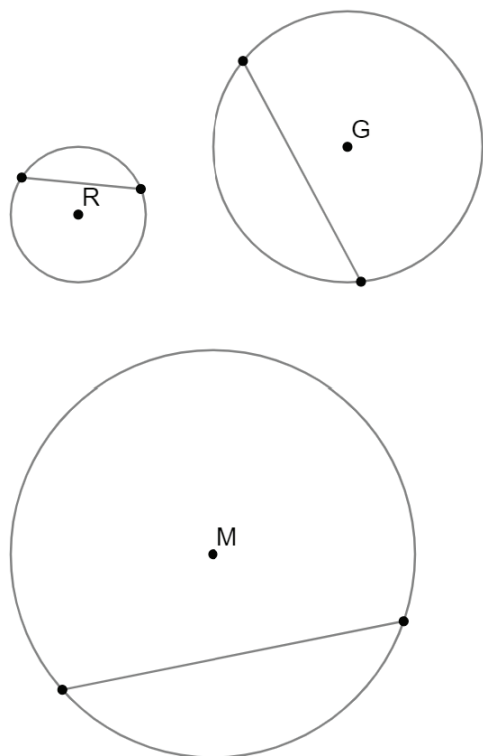


- a. What do you wonder?
- b. What do you notice?
- c. Write a question about the pyramid that a 5th grader could answer.



12.1.2 Exploration: Chords

1. Circles R , G , and M are shown. Each circle has a chord. Use patty paper to copy the circles. Use the patty paper copy to construct a perpendicular bisector for each chord by folding.



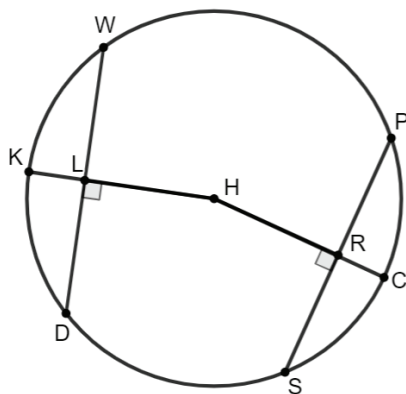
- a. Write a conjecture about the perpendicular bisector of a chord of a circle.

- b. Share your conjecture with your partner. Listen to your partner's conjecture and write a summary. Have your partner initial that it is summarized correctly.

Summary of Partner's Conjecture	Partner's Initials

- c. Review your conjecture and make any revisions, if needed.

2. Circle H is shown, where $\overline{KH}$ and $\overline{CH}$ are radii of the circle. $\overline{WD}$ and $\overline{PS}$ are chords of the circle equidistant from the center H . Therefore, $\overline{LH} \cong \overline{RH}$.

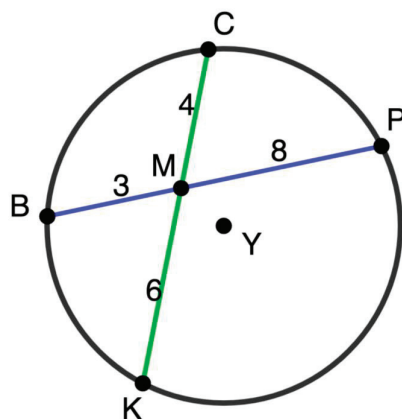


- a. Using your conjecture about perpendicular bisectors, complete each congruence statement.
- $\overline{DL} \cong$ _____
- $\overline{PR} \cong$ _____
- b. Draw a line segment from point D to point H and a line segment from point S to point H .
- c. Justify why $\overline{DH}$ and $\overline{SH}$ are congruent.
- d. Given that $\overline{LH} \cong \overline{RH}$ and $\overline{SH} \cong \overline{DH}$, what triangle congruence theorem justifies that $\triangle DLH$ and $\triangle SRH$ are congruent?

- e. Since $\triangle DLH \cong \triangle SRH$, what is true about $\overline{DL}$ and $\overline{SR}$?

- f. What is true about chords $\overline{DW}$ and $\overline{SP}$?

3. Circle Y is shown, where points C , P , K , and B are on circle Y . Chords $\overline{PB}$ and $\overline{KC}$ intersect at point M .

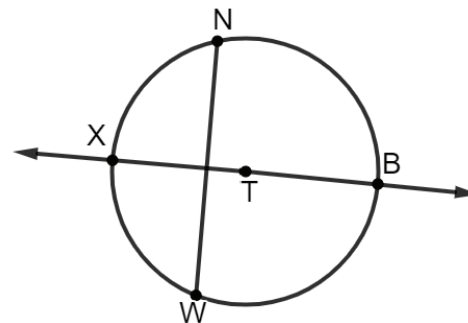


Observations of the Blue Chord Measures	Observations of the Green Chord Measures
Commonalities in the Blue and Green Chords	
What do you wonder about the relationship between the blue chord and the green chord?	



12.1.3 Bring It Together: Chord Relationships

1. Circle T is shown with chord $\overline{NW}$ and secant $\overleftrightarrow{XB}$, where $\overleftrightarrow{XB}$ bisects $\overline{NW}$ and $\overleftrightarrow{XB}$ is a diameter of the circle.

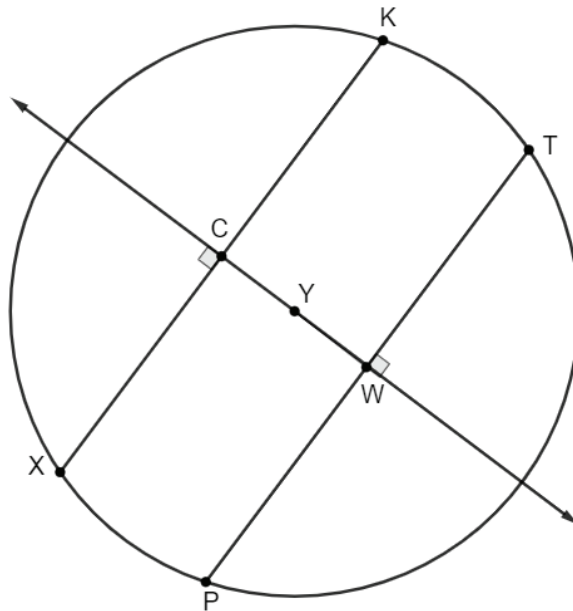


What information is known about circle T , chord $\overline{NW}$, and the angles formed by $\overleftrightarrow{XB}$ and $\overline{NW}$?



If a diameter of a circle is perpendicular to a chord, then the diameter bisects the chord and its arc.

2. Circle Y is shown with chords $\overline{XK}$ and $\overline{PT}$, where $\overline{CY} \cong \overline{WY}$.

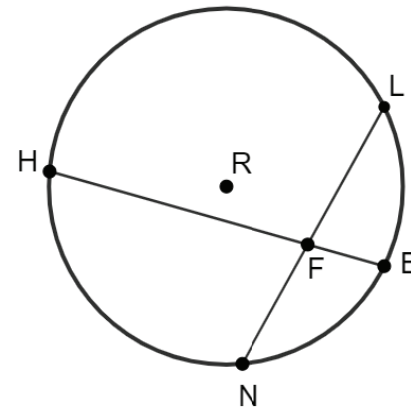


What information is known about circle Y , chords $\overline{XK}$ and $\overline{PT}$, and the angles formed by $\overleftrightarrow{CW}$?



If two chords are equidistant from the center of a circle, then the chords are congruent.

3. Circle R is shown with chords $\overline{HB}$ and $\overline{LN}$ that intersect at point F .



What information is known about circle R , chords $\overline{HB}$ and $\overline{LN}$, and the angles formed?

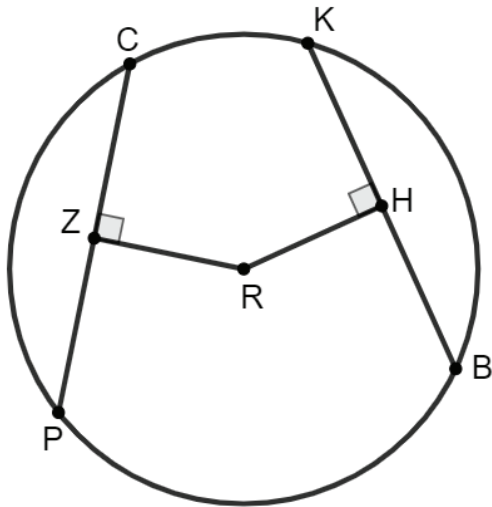


When two chords intersect each other inside a circle, the products of their segments are equal.

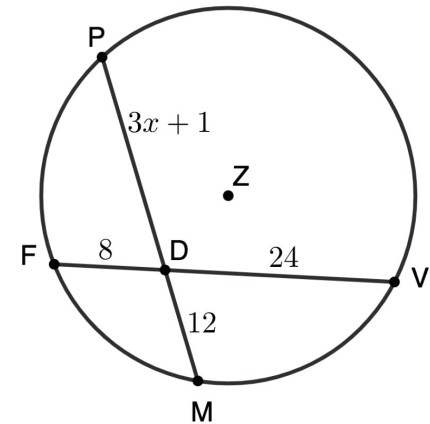


12.1.4 Guided Practice: Solving Problems Involving Intersecting Chords

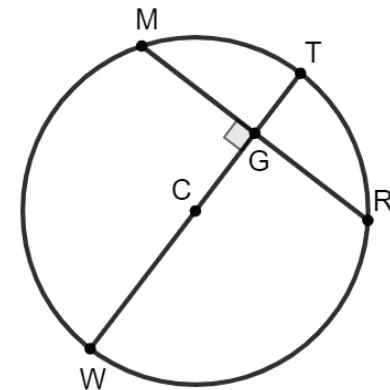
1. Circle R with points C , K , B , and P on the circle are shown with chords $\overline{CP}$ and $\overline{KB}$ equidistant from the center. Find the value of x , if $CP = 6x - 7$, $KB = 35$, $ZR = HR$, and $m\angle CZR = m\angle KHR = 90^\circ$.



2. In circle Z , chords $\overline{PM}$ and $\overline{FV}$ intersect at point D . Points P , F , M , and V lie on circle Z . Given $FD = 8$, $DM = 12$, $VD = 24$, and $DP = 3x + 1$, solve for x .



3. Circle C is shown, where points T , R , W , and M are on the circle. The diameter, $\overline{WT}$, is a perpendicular bisector of chord $\overline{MR}$. If $MG = y + 4$ and $GR = 2y + 1$, what is the length of the chord MR ?





12.1.5 Wrap-Up: Reflection

Draw an **X** on each line to indicate how you feel about solving problems involving chords.

1. I can use the proportional relationship between the measures of segments formed by intersecting chords to solve problems.

I need help. I can't get started. I'm getting there, but I need more practice. I understand this and I feel confident.



2. I can demonstrate that chords are congruent, if the two chords are equidistant from the center of a circle.

I need help. I can't get started. I'm getting there, but I need more practice. I understand this and I feel confident.



3. I can show that if a diameter of a circle is perpendicular to a chord, then the diameter bisects the chord.

I need help. I can't get started. I'm getting there, but I need more practice. I understand this and I feel confident.



Unit 12, Lesson 2: Solving Problems with Lengths of Two Secants



Benchmark

MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.



Learning Target

- ☐ I can solve mathematical and real-world problems that involve the lengths formed by the intersection of two secants.